UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

March 6, 2026

Mr. Christopher H. Mudrick Sr.
Senior Vice President
Constellation Energy Generation, LLC
President and Chief Nuclear Officer
Constellation Nuclear
200 Energy Way
Kennett Square, PA 19348

SUBJECT: BRAIDWOOD STATION, UNITS 1 AND 2; AND BYRON STATION, UNIT NOS. 1 AND 2 – ISSUANCE OF AMENDMENTS NOS. 247, 246, 244 AND 244 TO REVISE METHODOLOGY SUPPORTING THE USE OF FRAMATOME GAIA FUEL (EPID L-2025-LLA-0172)

Dear Mr. Mudrick:

The U.S. Nuclear Regulatory Commission (NRC or the Commission) has issued the enclosed amendments (listed below) in response to the Constellation Energy Generation, LLC (CEG) application dated November 12, 2025, as supplemented by letter dated February 24, 2026:

1. Amendment No. 247 to Renewed Facility Operating License No. NPF-72 and Amendment No. 246 to Renewed Facility Operating License No. NPF-77 for Braidwood Station, Units 1 and 2 (Braidwood), respectively;
2. Amendment No. 244 to Renewed Facility Operating License No. NPF-37 and Amendment No. 244 to Renewed Facility Operating License No. NPF-66 for Byron Station, Units Nos. 1 and 2 (Byron), respectively.

The proposed amendment would revise the methodology previously approved for Braidwood and Byron to allow loading of Framatome, Inc GAIA fuel with $M5_{Framatome}$ as a fuel cladding material.

The NRC staff has determined that the related safety evaluation (SE) contains proprietary information pursuant to Title 10 of the *Code of Federal Regulations* Section 2.390, "Public inspections, exemptions, request for withholding." The proprietary information is indicated by bold text enclosed with **[[double brackets]]**. The proprietary version of the SE is provided as enclosure 5. Accordingly, the NRC staff has also prepared a non-proprietary version of the SE, which is provided as enclosure 6.

Enclosure 5 to this letter contains proprietary information. When separated from Enclosure 5, this document is DECONTROLLED.

A copy of the NRC staff's Safety Evaluation is also enclosed. The Notice of Issuance will be included in the Commission's monthly *Federal Register* notice.

Sincerely,

/RA/

Scott P. Wall, Senior Project Manager
Plant Licensing Branch III
Division of Operating Reactor Licensing
Office of Nuclear Reactor Regulation

Docket Nos.: STN 50-456, STN 50-457,
STN 50-454, and STN 50-455

Enclosures:
1. Amendment No. 247 to NPF-72
2. Amendment No. 246 to NPF-77
3. Amendment No. 244 to NPF-37
4. Amendment No. 244 to NPF-66
5. Safety Evaluation (Proprietary)
6. Safety Evaluation (Non-Proprietary)

cc: Listserv

ADAMS Accession No.:
Package: ML26063A405
ML26063A419 (Non-Proprietary)
ML26063A418 (Proprietary)
e-Concurrence case: 20260304-60020

UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

CONSTELLATION ENERGY GENERATION, LLC

DOCKET NO. STN 50-456

BRAIDWOOD STATION, UNIT 1

AMENDMENT TO RENEWED FACILITY OPERATING LICENSE

Amendment No. 247
Renewed License No. NPF-72

1. The U.S. Nuclear Regulatory Commission (the Commission) has found that:

   A. The application for amendment by Constellation Energy Generation, LLC dated November 12, 2025, as supplemented by letter dated February 24, 2026, complies with the standards and requirements of the Atomic Energy Act of 1954, as amended (the Act), and the Commission's rules and regulations set forth in 10 CFR Chapter I;

   B. The facility will operate in conformity with the application, the provisions of the Act, and the rules and regulations of the Commission;

   C. There is reasonable assurance: (i) that the activities authorized by this amendment can be conducted without endangering the health and safety of the public, and (ii) that such activities will be conducted in compliance with the Commission's regulations;

   D. The issuance of this amendment will not be inimical to the common defense and security or to the health and safety of the public; and

   E. The issuance of this amendment is in accordance with 10 CFR Part 51 of the Commission's regulations and all applicable requirements have been satisfied.

2. Accordingly, by Amendment No. 247, Renewed Facility Operating License No. NPF-72 is hereby amended to authorize the revision to the Braidwood Final Safety Analysis Report as set forth in the application dated November 12, 2025, as supplemented by letter dated February 24, 2026, and as evaluated in the NRC staff's safety evaluation issued with this amendment.

3. This amendment is effective as of its date of issuance and shall be implemented prior to the Braidwood, Unit 2, Spring Refueling Outage 2026.

FOR THE NUCLEAR REGULATORY COMMISSION

ILKA BERRIOS

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Date: 2026.03.06 17:20:18 -05'00'

Ilka Berrios, Acting Chief
Plant Licensing Branch III
Division of Operating Reactor Licensing
Office of Nuclear Reactor Regulation

Attachment:
Changes to the Renewed Facility Operating License and Technical Specifications

Date of Issuance: March 6, 2026

UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

CONSTELLATION ENERGY GENERATION, LLC

DOCKET NO. STN 50-457

BRAIDWOOD STATION, UNIT 2

AMENDMENT TO RENEWED FACILITY OPERATING LICENSE

Amendment No. 246
Renewed License No. NPF-77

1. The U.S. Nuclear Regulatory Commission (the Commission) has found that:

   A. The application for amendment by Constellation Energy Generation, LLC dated November 12, 2025, as supplemented by letter dated February 24, 2026, complies with the standards and requirements of the Atomic Energy Act of 1954, as amended (the Act), and the Commission's rules and regulations set forth in 10 CFR Chapter I;

   B. The facility will operate in conformity with the application, the provisions of the Act, and the rules and regulations of the Commission;

   C. There is reasonable assurance: (i) that the activities authorized by this amendment can be conducted without endangering the health and safety of the public, and (ii) that such activities will be conducted in compliance with the Commission's regulations;

   D. The issuance of this amendment will not be inimical to the common defense and security or to the health and safety of the public; and

   E. The issuance of this amendment is in accordance with 10 CFR Part 51 of the Commission's regulations and all applicable requirements have been satisfied.

2. Accordingly, by Amendment No. 246, Renewed Facility Operating License No. NPF-77 is hereby amended to authorize the revision to the Braidwood Final Safety Analysis Report as set forth in the application dated November 12, 2025, as supplemented by letter dated February 24, 2026, and as evaluated in the NRC staff's safety evaluation issued with this amendment.

3. This amendment is effective as of its date of issuance and shall be implemented prior to the Spring Refueling Outage 2026.

FOR THE NUCLEAR REGULATORY COMMISSION

ILKA BERRIOS

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Date: 2026.03.06 17:20:41 -05'00'

Ilka Berrios, Acting Chief
Plant Licensing Branch III
Division of Operating Reactor Licensing
Office of Nuclear Reactor Regulation

Attachment:
Changes to the Renewed Facility
Operating License and
Technical Specifications

Date of Issuance: March 6, 2026

UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

CONSTELLATION ENERGY GENERATION, LLC

DOCKET NO. STN 50-454

BYRON STATION, UNIT NO. 1

AMENDMENT TO RENEWED FACILITY OPERATING LICENSE

Amendment No. 244
Renewed License No. NPF-37

1. The U.S. Nuclear Regulatory Commission (the Commission) has found that:

   A. The application for amendment by Constellation Energy Generation, LLC dated November 12, 2025, as supplemented by letter dated February 24, 2026, complies with the standards and requirements of the Atomic Energy Act of 1954, as amended (the Act), and the Commission's rules and regulations set forth in 10 CFR Chapter I;

   B. The facility will operate in conformity with the application, the provisions of the Act, and the rules and regulations of the Commission;

   C. There is reasonable assurance: (i) that the activities authorized by this amendment can be conducted without endangering the health and safety of the public, and (ii) that such activities will be conducted in compliance with the Commission's regulations;

   D. The issuance of this amendment will not be inimical to the common defense and security or to the health and safety of the public; and

   E. The issuance of this amendment is in accordance with 10 CFR Part 51 of the Commission's regulations and all applicable requirements have been satisfied.

2. Accordingly, by Amendment No. 244, Renewed Facility Operating License No. NPF-37 is hereby amended to authorize the revision to the Byron Final Safety Analysis Report as set forth in the application dated November 12, 2025, as supplemented by letter dated February 24, 2026, and as evaluated in the NRC staff's safety evaluation issued with this amendment.

3. This amendment is effective as of its date of issuance and shall be implemented prior to the Byron, Unit 2, Fall Refueling Outage 2026.

FOR THE NUCLEAR REGULATORY COMMISSION

ILKA BERRIOS

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Date: 2026.03.06 17:21:03 -05'00'

Ilka Berrios, Acting Chief
Plant Licensing Branch III
Division of Operating Reactor Licensing
Office of Nuclear Reactor Regulation

Attachment:
Changes to the Renewed Facility Operating License and Technical Specifications

Date of Issuance: March 6, 2026

UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

CONSTELLATION ENERGY GENERATION, LLC

DOCKET NO. STN 50-455

BYRON STATION, UNIT NO. 2

AMENDMENT TO RENEWED FACILITY OPERATING LICENSE

Amendment No. 244
Renewed License No. NPF-66

1. The U.S. Nuclear Regulatory Commission (the Commission) has found that:

   A. The application for amendment by Constellation Energy Generation, LLC dated November 12, 2025, as supplemented by letter dated February 24, 2026, complies with the standards and requirements of the Atomic Energy Act of 1954, as amended (the Act), and the Commission's rules and regulations set forth in 10 CFR Chapter I;

   B. The facility will operate in conformity with the application, the provisions of the Act, and the rules and regulations of the Commission;

   C. There is reasonable assurance: (i) that the activities authorized by this amendment can be conducted without endangering the health and safety of the public, and (ii) that such activities will be conducted in compliance with the Commission's regulations;

   D. The issuance of this amendment will not be inimical to the common defense and security or to the health and safety of the public; and

   E. The issuance of this amendment is in accordance with 10 CFR Part 51 of the Commission's regulations and all applicable requirements have been satisfied.

2. Accordingly, by Amendment No. 244, Renewed Facility Operating License No. NPF-66 is hereby amended to authorize the revision to the Byron Final Safety Analysis Report as set forth in the application dated November 12, 2025, as supplemented by letter dated February 24, 2026, and as evaluated in the NRC staff's safety evaluation issued with this amendment.

3. This amendment is effective as of its date of issuance and shall be implemented prior to the Fall Refueling Outage 2026.

FOR THE NUCLEAR REGULATORY COMMISSION

ILKA BERRIOS

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Date: 2026.03.06
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Ilka Berrios, Acting Chief
Plant Licensing Branch III
Division of Operating Reactor Licensing
Office of Nuclear Reactor Regulation

Attachment:
Changes to the Renewed Facility
Operating License and Technical
Specifications

Date of Issuance: March 6, 2026

UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

SAFETY EVALUATION BY THE OFFICE OF NUCLEAR REACTOR REGULATION

RELATED TO

AMENDMENT NO. 247 TO RENEWED FACILITY OPERATING LICENSE NO. NPF-72

AMENDMENT NO. 246 TO RENEWED FACILITY OPERATING LICENSE NO. NPF-77

AMENDMENT NO. 244 TO RENEWED FACILITY OPERATING LICENSE NO. NPF-37

AMENDMENT NO. 244 TO RENEWED FACILITY OPERATING LICENSE NO. NPF-66

CONSTELLATION ENERGY GENERATION, LLC

BRAIDWOOD STATION, UNITS 1 AND 2

BYRON STATION, UNIT NOS. 1 AND 2

DOCKET NOS. STN 50-456, STN 50-457, STN 50-454, AND STN 50-455

1.0 PROPOSED CHANGE

By letter dated November 12, 2025 (Agencywide Documents Access and Management System Package Accession No. ML25318B206), as supplemented by letter dated February 24, 2026 (ML26055A319), Constellation Energy Generation, LLC (CEG, or the licensee) submitted a license amendment request (LAR) to the U.S. Nuclear Regulatory Commission (NRC, the Commission) for Braidwood Station, Units 1 and 2 (Braidwood) and Byron Station, Unit Nos. 1 and 2 (Byron). The licensee requested revisions to methodology previously approved to allow the use of Framatome GAIA fuel at Braidwood and Byron.

From December 4, 2025, through February 28, 2026, the NRC staff conducted a regulatory audit to support its review of the amendment request, as discussed in the staff's audit plan dated December 15, 2025 (ML25342A113), and audit summary dated March 2, 2026 (ML26056A177).

The supplemental letter dated February 24, 2026, provided additional information that clarified the application, did not expand the scope of the application as originally noticed, and did not change the NRC staff's original proposed no significant hazards consideration determination as published in the *Federal Register* (FR) on December 30, 2025 (90 FR 61169).

2.0 REGULATORY EVALUATION

2.1 Background

On October 16, 2023, the NRC staff found the Framatome Topical Report ANP-10339P-A Revision 0, "ARITA – ARTEMIS/RELAP Integrated Transient Analysis Methodology" (ARITA) (Package, ML23233A151), acceptable for referencing in licensing applications for nuclear power plants. Section 5.0, "Applicability, Limitations, and Conditions" (ML23299A232), of the staff's safety evaluation (SE), outlines applicability limits Framatome proposed for the ARITA methodology, as well as limitations and conditions (L&Cs) that the NRC staff found necessary to support its review conclusions. Specifically,

- L&C 3 requires that plant submittals include information concerning the analyzed figures-of-merits (FOMs), as well as the defined sampling ranges for key parameters **[[** **]]** and the means by which they were determined.

- L&C 10 addresses **[[** **]]** and requires this parameter **[[** **]]**, unless justified otherwise.

On August 21, 2025 (ML25153A008), the NRC issued amendments revising the Braidwood and Byron technical specifications (TSs) to allow loading of Framatome, Inc., GAIA fuel with $M5_{Framatome}$ as a fuel cladding material. In part, these amendments added ANP-10339P-A Revision 0, "ARITA – ARTEMIS/RELAP Integrated Transient Analysis Methodology," October 2023, to TS 5.6.5, "Core Operating Limits Report (COLR)", as an analytical method allowed to be used to determine the core operating limits. As part of the basis for approving these amendments, L&Cs 3 and 10 are to be used by the following method:

- L&C 3: The FOMs for each event analyzed were provided by the licensee. Further, [[**]]** that was consistent with the ARITA methodology.

- L&C 10: **[[** **]]**

In the ARITA methodology, the uncertainty of **[[** **]]**. Using the ARITA methodology approved in the August 21, 2025, amendments **[[** **]]**

2.2 Proposed Changes

The licensee's proposed changes to how the ARITA methodology is applied at Braidwood and Byron. Specifically, CEG is proposing changes to L&C 10 regarding the sampling range **[[** **]]** and to L&C 3 regarding Braidwood and Byron key parameter sampling strategy.

2.2.1 Original Requested Change

In the November 12, 2025, application, CEG requested to modify L&Cs 3 and 10 with the following method:

- L&C 3:
  - **[[** **]]**
  - **[[** **]]**

    NOTE: Analyses which **[[** **]]** using the previously approved methods remains valid. **[[** **]]** the ARITA methodology.

- L&C 10:
  - **[[** **]]** in all applicable safety analyses. **[[** **]]** in ARITA-based analyses **[[** **]]** This proposed change applies to all transients evaluated using the ARITA methodology **[[** **]]** including those supporting Departure from Nucleate Boiling (DNB), Fuel Centerline Melt (FCM), and Transient Cladding Strain (TCS) evaluations**.** **[[** **]]**

2.2.2 Updated Requested Change

In the February 24, 2026, supplement, CEG changed the proposed methodology to modify L&C 10 with the following method:

- L&C 3: The supplement did not modify L&C 3.
  - **[[**

    ]]
  - **[[**

    ]]

    NOTE: Analyses which **[[** **]]** using the previously approved methods remains valid. **[[**

    ]] the ARITA methodology.
- L&C 10:
  - **[[**

    ]] in all applicable safety analyses. The **[[** **]]** within the ARITA methodology (i.e. all transients evaluating using the ARITA methodology that rely on this correlation) are proposed to be updated as follows: **[[**

    ]]

2.3 Applicable Regulations and Guidance

The NRC staff considered the following regulatory requirements and guidance during its review of the LAR.

Regulatory Requirements

The regulations in 10 CFR 50.90 state that whenever a holder of an operating license desires to amend the license, including TSs in the license, an application for amendment must be filed with the Commission fully describing the changes desired. The regulations at 10 CFR 50.92(a) state that determinations on whether to grant an applied for license amendment are guided by the considerations that govern the issuance of initial licenses to the extent applicable and appropriate. Both the common standards for licenses in 10 CFR 50.40(a) (regarding, among

other things, consideration of the operating procedures, the facility and equipment, the use of the facility, and other TSs, or the proposals) and those specifically for issuance of operating licenses in 10 CFR 50.57(a)(3), provide that there must be reasonable assurance that the activities at issue will not endanger the health and safety of the public, and that the applicant will comply with the Commission's regulations.

The regulations under 10 CFR 50.36, "Technical specifications," provide regulatory requirements related to the content of TSs. Section 50.36(b) of 10 CFR requires that each license authorizing the operation of a facility will include TSs and that the TSs will be derived from the safety analysis. Section 50.36(c) of 10 CFR specifies the categories that are to be included in the TSs including (1) safety limits, limiting safety system settings, and limiting control settings; (2) LCOs; (3) SRs; (4) design features; and (5) administrative controls.

Regulations in 10 CFR Part 50, Appendix A, General Design Criteria (GDC), Criterion 10 (GDC 10), "Reactor design," state that "[t]he reactor core and associated coolant, control, and protection systems shall be designed with appropriate margin to assure that specified acceptable fuel design limits are not exceeded during any condition of normal operation, including the effects of anticipated operational occurrences."

Regulatory Guidance

The NRC staff relied on the following sections of NUREG-0800, "Standard Review Plan (SRP) for the Review of Safety Analysis Reports for Nuclear Power Plants: LWR Edition" (SRP) in its review of this LAR:

- Chapter 15, "Introduction – Transient and Accident Analyses," Revision 3, dated March 2007 (ML070710376).
- Section 15.0.2, "Review of Transient and Accident Analysis Methods," Revision 0, dated March 2007 (ML070820123).

Regulatory Guide (RG) 1.203, "Transient and Accident Analysis Methods," dated December 2005 (ML053500170), describes a process for developing and assessing evaluation models that is known as the evaluation model development and assessment process (EMDAP). While the sequence and formatting differ, the EMDAP of RG 1.203 is consistent with the guidance provided in Section 15.0.2 of the SRP.

3.0 TECHNICAL EVALUATION

The NRC staff evaluated the licensee's application to determine whether the proposed changes to L&C 3 and L&C 10 of the ARITA methodology are consistent with the regulations, guidance, and plant-specific design and licensing basis discussed in Section 2.3 of this SE. The NRC staff reviewed the licensee's statements in the LAR, as supplemented, and the relevant sections of the Braidwood and Byron TS and Updated Final Safety Analysis Report (UFSAR). The staff reviewed the changes to L&Cs 3 and 10 for the ARITA methodology to ensure they are met satisfactorily.

3.1 Limitation and Condition 3

Limitation and Condition 3 of the ARITA methodology requires that plant submittals include information concerning the analyzed FOMs as well as **[[** **]]** and the means by which the key parameters were determined. As discussed in the NRC staff's SE for the ARITA methodology, the **[[** **]]** analyzed FOMs. In Section 2.2 of the present application, the license proposed an alternative method for **[[** **]]** indicating the proposed approach is consistent with L&C 3. The proposal consists of the original approach with two key modifications:

1. **[[** **]]**
2. **[[** **]]**

These changes apply to the **[** **]]** the ARITA methodology.

With regard to the first of these modifications, the NRC staff's position has been that **[[** **]]** The licensee's assertion is that the **[[** **]]** The licensee further asserts that, in some instances, **[[** **]]** As justification of these assertions, the licensee effectively delineates three separate classifications of **[[** **]]** The three classifications are **[[** **]]**

For **[[** **]]** the licensee indicates the impact on pertinent FOMs **[[** **]]** As an example, the licensee provided plots demonstrating the variation in **[[** **]]** The results showed the variation in **[[** **]]** These comparisons were specifically chosen because of the *a priori* known significant impact **[[** **]]**

However, for some events and FOMs, **[[**
]] In such instances, **[[**
]] Per the response to request for additional information (RAI)-11 of ANP-10339P-A, Revision 0, the **[[** **]]** The licensee classifies these **[[**

]] The licensee indicates that **[[**

]]

Although not directly stated in the LAR, the NRC staff notes the licensee's justification that **[[**
]] rests on the identification of trends. The licensee refers to **[[**
]] This noise detracts from the clear identification of the underlying trends that **[[**
]] This is demonstrated in the example plots provided by the licensee, wherein a clearly discernable trend leads to **[[**
]] While multiple **[[**
]] the NRC staff concluded it is unlikely a **[[**
]] and highly unlikely that it will produce **[[**

]]

With respect to **[[** **]]** per the response to RAI-11 of ANP-10339P-A, Revision 0, **[[**

]]

The licensee does not provide any hard criteria to delineate when a **[[**
]] However, because the **[[**

]] the NRC staff finds that this is reasonable. **[[**

]] As discussed above, **[[**

]]

The NRC staff notes that for **[[**

]] As noted by the NRC staff in discussions with Framatome during the review of ANP-10339P-A, Revision 0, as alluded to in the NRC staff's SE for ANP-10339P-A, Revision 0, and as observed in audited analyses reviewed for the August 21, 2025 amendment, the **[[**

]] The NRC staff expects this practice to continue.

Based on the above discussion, the NRC staff finds the first modification to L&C 3 to be acceptable.

With regard to the second of the modifications, the NRC staff examined the **[[**
]] presented in audit documentation. It was not the intent of the NRC staff as part of the review of ANP-10339P-A, Revision 0, to prescribe or approve a detailed generic process for performing the **[[**
]]
Rather, the scope presented in the NRC staff's SE to ANP-10339P-A, Revision 0**, [[**
]] was intended to allow for flexibility. Similarly, it is not the intent of the NRC staff within the present SE to generically approve the process for and **[[**
]] that is presented in the audited documentation. Determining the **[[** **]]** is expected to require engineering judgement and assessment, even as additional experience is gained in the application of the ARITA methodology. The process presented in the audited documentation leverages engineering judgement **[[**
]] and it is expected that the process will undergo continual refinement.

Therefore, the NRC staff broadly assessed the consistency with which the process meets the intent of the original L&C 3. The staff concludes the process is reasonable because it utilizes and follows the key steps outlined in the SE for ANP-10339P-A, Revision 0, while providing the information required in the original L&C 3. The staff also concludes **[[**

]] is not unreasonable provided engineering judgement is continually leveraged, particularly when new core designs are introduced as these are expected to result in different cycle-to-cycle deviations.

Based on the above discussion, the NRC staff finds the second modification to L&C 3 to be acceptable.

The NRC staff noted the licensee stated that analyses which **[[** **]]** using the previously approved methods remain valid. The staff finds this is reasonable because the previous analyses utilized an approved method and the proposed modifications introduced in the present LAR seek to refine the method to **[[** **]]** not correct or compensate for errors or non-conservatisms.

3.2 Limitation and Condition 10

As discussed in Section 2.1 of this SE, L&C 10 addresses **[[** **]]**, unless justified otherwise. In the August 21, 2025, amendment, the NRC staff approved a **[[** **]]**

In Section 2.1 of the present application, the licensee stated that **[[** **]]** As such, the licensee proposed a change that **[[** **]]** in the ARITA-based analyses and **[[** **]]** in the ARITA uncertainty analysis. This proposed change would apply to all transients evaluated using the ARITA methodology **[[** **]]**

The justification provided for the proposed approach consists of two parts. First, the NRC-prescribed **[[** **]]** in L&C 10 represents laboratory-scale data scatter in the correlation and not the **[[** **]]**. Second, the steam generator heat transfer models are explicitly calibrated to plant data to reproduced measured steady-state conditions prior to initiating transient analyses, and this calibration substantially offsets bias and minimizes scatter from the embedded heat transfer correlations. Any remaining residual uncertainty would be due to uncertainty in the calibrated domain and extrapolation uncertainty associated with departures from that domain during transients. Such uncertainties would be exceedingly small and not have a significant impact on transient progression for events considered within the ARITA methodology.

In assessing the second part of the justification, the NRC staff examined the steam generator nodalization and initialization process as described within the approved ARITA methodology, including the heat transfer correlations implemented, the **[[** **]]** during steam generator initialization to steady-state plant data. Depending on the event being analyzed, different numbers of steam generator nodes will enter into and out of different heat transfer regimes, switching from one applicable heat transfer correlation to another, and thus

impacting the overall dominant heat transfer mechanism for the steam generator. This is typical and expected.

However, the NRC staff noted the **[[** **]]** While the dominant heat transfer mechanism in the steam generator (e.g., subcooled convection or nucleate boiling) may be the same for multiple steady state conditions, **[[** **]]** Thus, while it is possible the uncertainty **[[** **]]** it is less likely the uncertainty **[[** **]]**

As a result of the above assessment, the NRC staff identified a number of questions and concerns that would require resolution before a determination could be reached regarding the acceptability of the licensee's justification **[[** **]]** However, the NRC staff concluded the time necessary to pursue resolution of the questions and concerns would exceed the allotted review timeframe. In a virtual audit conducted between December 4, 2025, through February 28, 2026, the NRC staff discussed this conclusion with the licensee and Framatome. The licensee and Framatome ultimately requested the NRC staff focus its efforts on the first part of the justification **[[** **]]** Thus, the NRC staff made no regulatory conclusions regarding the proposal to **[[** **]]**

Regarding the first part of the justification and the acceptability of **[[** **]]** Section 3.5.2.1 of the staff's SE for the ARITA methodology discusses **[[** **]]** an average deviation consistent with NRC staff experience and reported in sources such as text books and early journal articles **[[** **]]** However, more recent-to-contemporary explorations in the scientific literature regarding the performance of **[[** **]]** identify the uncertainty is much less than **[[** **]]** when the available data is used to fully characterized a statistical predicted-versus-measured distribution. Notably, **[[** **]]** Additionally, experimental data collected by **[[** **]]** the 95 percent tolerance limits further shrink to **[[** **]]** suggesting a one standard deviation uncertainty of approximately **[[** **]]**

Because the fit of a correlation is data-driven, the **[[**

]] correlation implemented within ARITA. However, the NRC staff noted that both articles from the literature, when compared against each other and earlier sources, demonstrate that, as the rigor and precision with which experimental data is collected, the spread in the predicted-versus-measured results decreases, suggesting the originally reported average deviation is due in no small part to large experimental uncertainties. Further, comparisons between the **[[** **]]** results demonstrate that, for the **[[** **]]** ranges applicable to pressurized water reactor (PWR) steam generator performance, the correlations perform substantially the same for the newly acquired data. These observations indicate the **[[** **]]** 95 percent tolerance limits and resulting one standard deviation uncertainty reasonably characterize the **[[** **]]** over the ranges of conditions in which the correlation is applied within the ARITA methodology.

In support of the requested uncertainty reduction to **[[** **]]** the licensee referenced a journal article authored by **[[** **]]** This article examines the performance of an array of heat transfer correlations to the data available for various fluids at supercritical pressures for purposes of validating the correlations for supercritical water-cooled reactors. This article indicates that a direct, unmodified application of the **[[** **]]** to the available data for water yields a predicted-to-measured uncertainty of approximately **[[** **]]** While the correlation is applied unmodified, the NRC staff notes the database represents water at super critical conditions, which is not characteristic of conventional PWRs. At supercritical conditions, the heat transfer to the fluid can change drastically, potentially increasing before experiencing a reduction as fluid conditions progress further into the supercritical domain (i.e., pressure increases beyond the supercritical point for a given temperature).

Framatome indicated in ANP-4184, Revision 0, Supplement 1P, Revision 0, that for single-phase forced convection, pressure does not appear explicitly in the **[[**

]] Framatome also indicated that ultimately, the proposed magnitude in reduction of the uncertainty should not be interpreted as a precise statistical descriptor derived from a single experimental database. Rather, it represents a conservative characterization of the variability associated with a steam generator heat transfer model that has been conditioned (calibrated) to plant performance.

In assessing the applicability of the supercritical data, the NRC staff noted the results were derived from a substantial database, in excess of 20,000 data points. The vast majority of these datapoints exist at pressures beyond the supercritical point, where heat transfer to the fluid experiences a reduction, returning closer to non-supercritical performance (i.e., behavior approaches that of typical single-phase turbulent convection). Further, various experiments from which the data were collected appear to be characterized by an increased level of rigor and precision in the collection process compared to earlier sources. This is consistent with staff observations noted above between data collected by **[[** **]]** which suggests a minor reduction in uncertainty compared to previously reported results would not be unusual.

Based on the aggregate of the assessment points discussed above, and with consideration that, as discussed previously, it is possible the uncertainty **[[**

]] it is the engineering judgement of the NRC staff that, for the present application of the ARITA methodology with steam generator nodalization and initialization as defined in ANP-10339P-A, Revision 0, a reduction in the uncertainty magnitude for **[[** **]]** correlation is acceptable and that the value of **[[** **]]** for one standard deviation is reasonable. The uncertainty sampling will continue to be performed over a range of **[[** **]]**, which is consistent with the range approved in the August 2025 amendment. The staff notes that this change does not invalidate the analyses performed in that amendment or the analyses performed to-date for Byron and Braidwood in support of the present amendment; in comparison to the range of possible heat transfer coefficients obtained when sampling with the reduced one standard deviation uncertainty magnitude of **[[** **]]**, the existing analyses were performed utilizing a much wider range of sampled values that will yield a more conservative estimate of the 95/95 tolerance limit on the figures of merit.

3.3 Technical Conclusion

The NRC staff has reviewed the LAR and its supplement to evaluate the acceptability of the proposed modifications to limitation and condition 3 and 10 of the ARITA methodology for Braidwood and Byron transition to Framatome GAIA fuel design with Framatome safety analyses and core design methodologies. Based on its review, the NRC staff has determined that the licensee provided an adequate technical basis to support the proposed LAR. Specifically, the NRC staff has determined that the licensee has demonstrated (1) that the proposed modifications of (a) **[[** **]]** for limitation and condition 3, and (b) **[[** **]]** for limitation and condition 10, are technically acceptable and consistent with the ARITA methodology used in the execution of the LAR, (2) Framatome codes and methods are applicable for Braidwood and Byron, and (3) the safety analysis results submitted to the NRC staff demonstrate compliance with applicable regulatory requirements. Therefore, the NRC staff finds there is reasonable assurance of public health and safety.

4.0 STATE CONSULTATION

In accordance with the Commission's regulations, the Illinois State official was notified of the proposed issuance of the amendments on February 12, 2026. The State official had no comments.

5.0 ENVIRONMENTAL CONSIDERATION

The amendment changes a requirement with respect to installation or use of a facility component located within the restricted area as defined in 10 CFR, Part 20. The NRC staff has determined that the amendment involves no significant increase in the amounts, and no significant change in the types of any effluents that may be released offsite, and that there is no significant increase in individual or cumulative occupational radiation exposure. The Commission has previously issued a proposed finding that the amendment involves no significant hazards consideration, which was published in the *Federal Register* on December 30, 2025 (90 FR 61169), and there has been no public comment on such finding. Accordingly, the amendment meets the eligibility criteria for categorical exclusion set forth in 10 CFR 51.22(c)(9). Pursuant to 10 CFR 51.22(b) no environmental impact statement or environmental assessment need be prepared in connection with the issuance of the amendment.

6.0 CONCLUSION

The Commission has concluded, based on the considerations discussed above, that: (1) there is reasonable assurance that the health and safety of the public will not be endangered by operation in the proposed manner, (2) there is reasonable assurance that such activities will be conducted in compliance with the Commission's regulations, and (3) the issuance of the amendment will not be inimical to the common defense and security or to the health and safety of the public.

7.0 REFERENCES

1) McAdams WH (1942) "Heat transmissions," 2nd edition. McGraw-Hill, New York.
2) **[[**

3)

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4) Hussam A. M. Zahlan, Laurence K. H. Leung, Yan-Ping Huang, and Guang-Xu Liu, "Assessment of Convective Heat Transfer Correlations Against an Expanded Database for Different Fluids at Supercritical Pressures", *ASME Journal of Nuclear Engineering and Radiation Science,* Vol. 4, Issue. 1, pp. 011004-1 (2018). https://doi.org/10.1115/1.4037720

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